SYNTHETIC VALIDATION − SuSiE versus EM SuSiE−mix | iteration 18

113 eligible gene−tissue regions across 6 tissues | tentative four−panel comparison

A Where do CS counts agree?

Rows: SuSiE CSs | Columns: EM CSs | Cells: gene−tissue regions

Tissue1
Most disagreement | n = 20
Same: 4 (20.0%) | Different: 16

	0	1	2
0	−	0	0
1	0	4	16
2	0	0	0

Tissue2
Most disagreement | n = 20
Same: 8 (40.0%) | Different: 12

	0	1	2
0	−	0	0
1	0	8	12
2	0	0	0

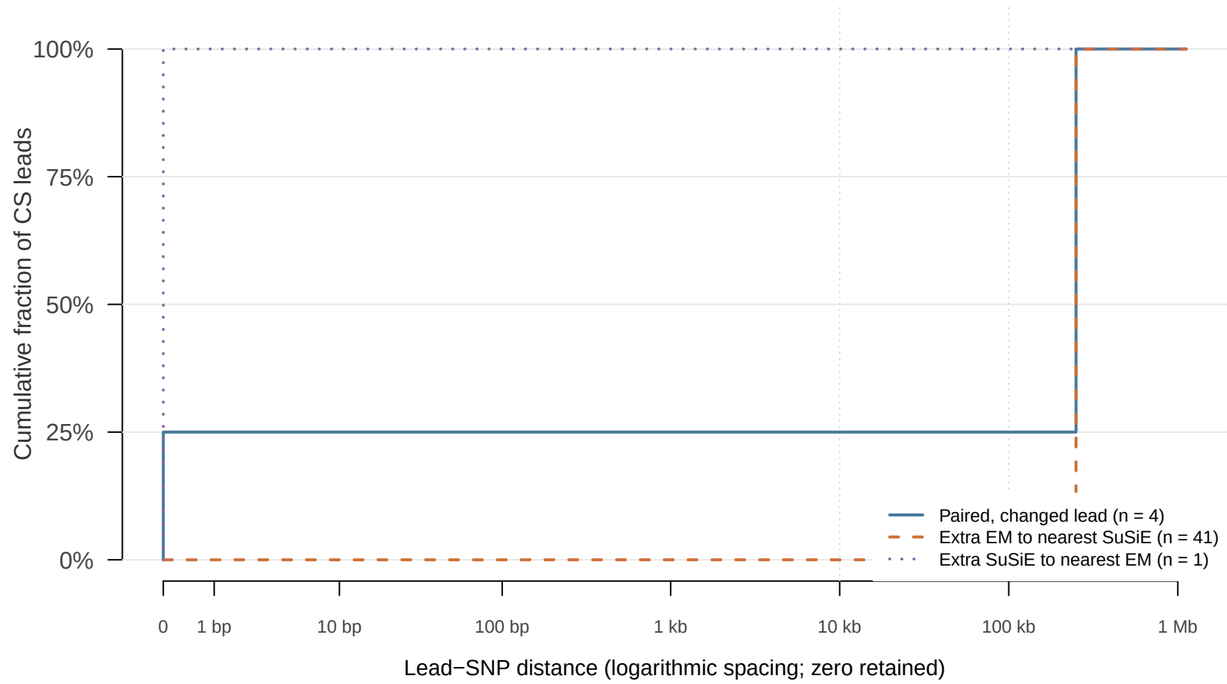
Tissue5
Most agreement | n = 20
Same: 20 (100.0%) | Different: 0

	0	1
0	−	0
1	0	20

Tissue4
Most agreement | n = 20
Same: 16 (80.0%) | Different: 4

	0	1	2
0	−	0	0
1	0	16	4
2	0	0	0

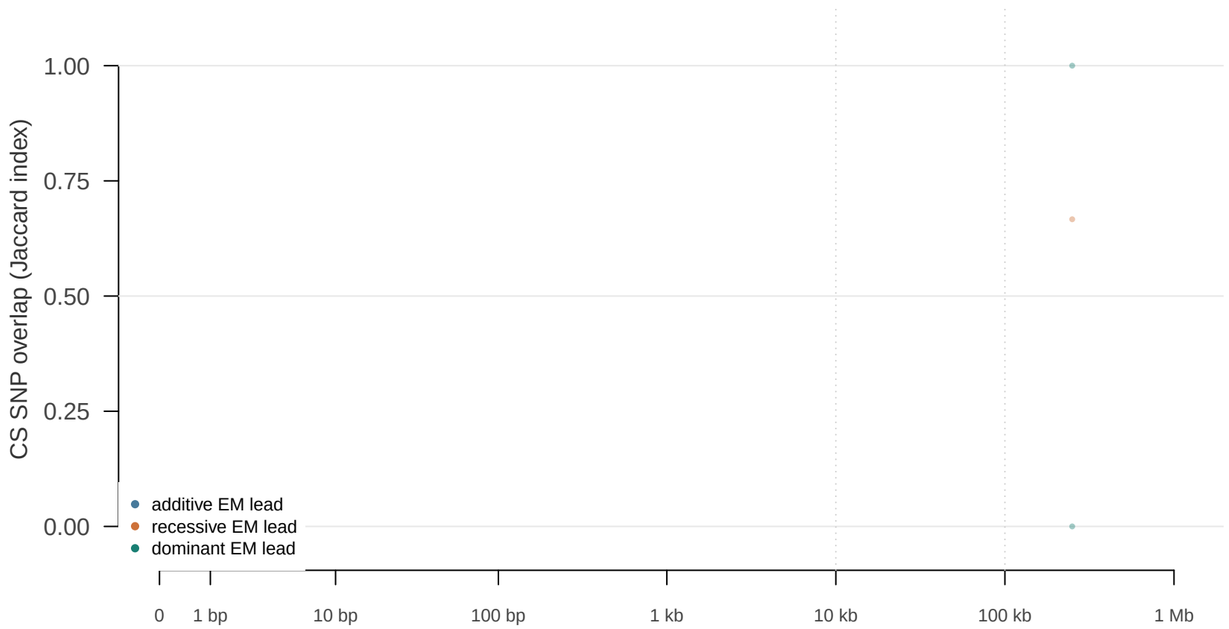
B Lead retention and displacement: all tissues



Not in curves: 106 identical pairs; 2 extras with no other−model CS

C Does a shifted lead change the credible set?

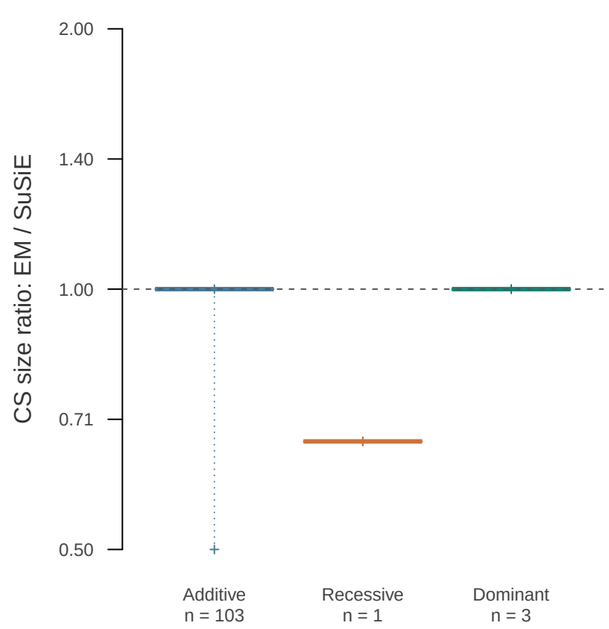
Changed pairs: 4 | identical sets: 1 | disjoint sets: 2



Jaccard: 0 = disjoint; 1 = identical SNP sets. Missing membership: 0 pairs.

D Resolution and support

Matched CSs, grouped by EM lead coding

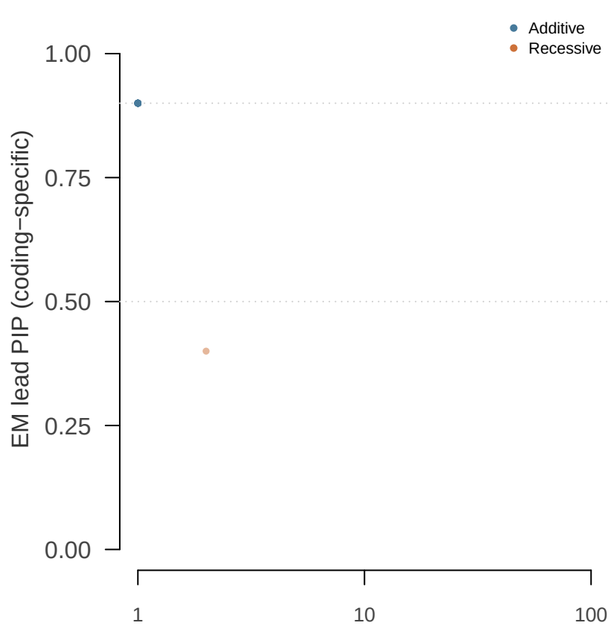


Below 1: fewer SNPs in the EM CS

Missing size: 3 pairs | whiskers: 1.5 IQR; +: range

Additional EM CSs

n = 42 | PIP >= 0.9: 41



Unavailable size/PIP: 0; no SuSiE CS: 1

SNP sets collapse genetic codings. Pairing does not establish biological equivalence; smaller sets do not establish better coverage.

Pairing reserves identical lead IDs, then minimizes total distance without reusing CSs. Additional = unmatched under this rule.

Cohort: P < 1e−08; mean reads >= 100; at least one CS in either model. Tissue ranking requires n >= 15.